

Nils Lukas

Assistant Professor • MBZUAI • Abu Dhabi, UAE

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Updated on July 26, 2026

Research Interests

Design secure and private AI agents in the presence of untrustworthy **providers**, **data**, **models**, and **users**.

1. **Providers**: Confidential computing & encrypted inference.
2. **Data**: Poisoning, prompt injection & training-data privacy.
3. **Models**: Scalable oversight & differentially private inference.
4. **Users**: Detect misuse via watermarking & safety monitoring.

Research Grants

Awarded	\$1,458,000
[United AI-Saqer Group] MedBoard: Privacy-preserving Multi-Agent AI for Healthcare	\$136,000 · 2026
PI: Nils Lukas · Co-PIs: Praneeth Vepakomma , Preslav Nakov	
[IAAI Research Program] Trust Layer for Safe Smallholder Advisory	\$500,000 · 2026
PI: Salem Lahlou · Co-PIs: Nils Lukas , Martin Takáč	
[Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments	\$100,000 · 2025
PI: Nils Lukas · Co-PI: Praneeth Vepakomma	
[Etihad Airways] Conversational Booking Agents	\$450,000 · 2025
PI: Nils Lukas · Co-PIs: Salem Lahlou , Alham Fikri Aji , Martin Takáč , Mingming Gong	
[United AI-Saqer Group] Privacy-preserving Brain–Computer Interfaces	\$136,000 · 2025
PI: Abdulrahman Mahmoud · Co-PIs: Nils Lukas , Elizabeth Churchill	
[TII Funding] GFlowNets for Fuzzing of Agentic Applications	\$136,000 · 2025
PIs: Salem Lahlou & Nils Lukas	

Conference Publications

Underlined authors are students and postdocs I supervise as primary advisor.

27	Entropy-Gated Latent Recursion
[NeurIPS'26]	Soham Bhattacharjee , Dushyant Singh Chauhan , Salem Lahlou , Martin Takáč , Nils Lukas . The Fortieth Annual Conference on Neural Information Processing Systems, 2026.
26	PROBE: Learning to Audit Policy Compliance in Tool-Using LLM Agents
[NeurIPS'26]	Kshitij Mishra , Abhijith Sharma , Nils Lukas , Salem Lahlou . The Fortieth Annual Conference on Neural Information Processing Systems, 2026.
25	CopyShield: A Cross-Level Benchmark of Copyright Defenses in LLMs
[EMNLP'26]	Maryam Alshehryari , Dushyant Chauhan , Samuele Poppi , Martin Takáč , Salem Lahlou , Nils Lukas . Conference on Empirical Methods in Natural Language Processing (Main Conference), 2026.
Main Conference AR: 15.4% (2,719/17,669)	
🏆 Oral Presentation	
24	TRACER: Trace-Regularized Training Against Indirect Prompt Injection
[EMNLP'26]	Kshitij Mishra , Salem Lahlou , Nils Lukas . Findings of the Association for Computational Linguistics: EMNLP, 2026.
Findings AR: 14.3% (2,533/17,669)	

23	SPQR: A Multi-Dimensional Benchmark for Safety Alignment under Benign Model Adaptation Mohammed Talha Alam, Nada Saadi, Fahad Shamshad, Nils Lukas , Karthik Nandakumar, Fakhri Karray, <u>Samuele Poppi</u> . European Conference on Computer Vision, 2026.
22	RAVEN: Erasing Invisible Watermarks via Novel View Synthesis Fahad Shamshad, Nils Lukas , Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
[CVPR'26] AR: 25.4% (4,090/16,092) 🏆 Oral (top 0.8%)	
21	Collaborative Threshold Watermarking <u>Tameem Bakr</u> , Anish Ambreth, Nils Lukas . The Forty-Third International Conference on Machine Learning, 2026.
[ICML'26] AR: 26.6% (6,352/23,918)	
20	CoRe: Collaborative Reasoning via Cross Teaching <u>Kshitij Mishra</u> , Mirat Aubakirov, Martin Takáč, Nils Lukas , Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.
[ICML'26] AR: 26.6% (6,352/23,918)	
19	Forest Before Trees: Latent Superposition for Efficient Visual Reasoning Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, Nils Lukas , <u>Yuhan Liu</u> . The 64th Annual Meeting of the Association for Computational Linguistics, 2026.
[ACL'26] Main Conference AR: 18.9% (2,296/12,148)	
18	Detecting AI-Generated Video: A Vision-Language Dual-View Survey Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, Nils Lukas , Gus Xia, Xiuying Chen, <u>Yuhan Liu</u> . Findings of the Association for Computational Linguistics, 2026.
[ACL'26] Findings AR: 17.8% (2,163/12,148)	
17	DP-Fusion: Token-Level Differentially Private Inference for Large Language Models <u>Rushil Thareja</u> , Preslav Nakov, <u>Praneeth Vepakomma</u> , Nils Lukas . The Fourteenth International Conference on Learning Representations, 2026.
[ICLR'26] AR: 27.4% (5,355/19,525)	
16	SD-E2: Semantic Exploration for Reasoning Under Token Budgets <u>Kshitij Mishra</u> , Nils Lukas , Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.
[EACL'26] Findings	
15	Mask Image Watermarking Runyi Hu, Jie Zhang, Shiqian Zhao, Nils Lukas , Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.
[NeurIPS'25] AR: 24.5% (5,290/21,575)	
14	SPIRIT: Patching Speech Language Models against Jailbreak Attacks Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, Nils Lukas . Empirical Methods in Natural Language Processing (Main Conference), 2025.
[EMNLP'25] AR: 22.2% (1,811/8,172)	
13	Optimizing Adaptive Attacks against Content Watermarks for Language Models Abdulrahman Diaa, <u>Toluwani Aremu</u> , Nils Lukas . The Forty-Second International Conference on Machine Learning, 2025.
[ICML'25] AR: 26.9% (3,260/12,107) ★ Spotlight (Top 2.6%)	
12	Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, Nils Lukas , Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.
[ICML'25] AR: 26.9% (3,260/12,107)	

- 11
[USENIX'24]
AR: 19.1%
(417/2,176)

PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting
Rasoul Mahdavi, **Nils Lukas**, Faezeh Ebrahimiaghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
- 10
[USENIX'24]
AR: 19.1%
(417/2,176)

Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimiaghazani, Bailey Kacsmar, Xinda Li, **Nils Lukas**, Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
- 9
[ICLR'24]
AR: 30.8%
(2,250/7,262)

Leveraging Optimization for Adaptive Attacks on Image Watermarks
Nils Lukas, Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
- 8
[ICLR'24]
AR: 30.8%
(2,250/7,262)
● Media Coverage

Universal Backdoor Attacks
Benjamin Schneider, **Nils Lukas**, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
- 7
[USENIX'23]
AR: 29.2%
(422/1,444)

PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators
Nils Lukas and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.
- 6
[S&P'23]
AR: 17.0%
(195/1,147)
★ Distinguished Contribution Award at Microsoft MLADS

Analyzing Leakage of Personally Identifiable Information in Language Models
Nils Lukas, Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.
- 5
[S&P'22]
AR: 14.5%
(147/1,012)

SoK: How Robust is Image Classification Deep Neural Network Watermarking?
Nils Lukas, Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.
- 4
[ICLR'21]
AR: 28.7%
(860/2,997)
★ Spotlight (Top 5%)

Deep Neural Network Fingerprinting by Conferrable Adversarial Examples
Nils Lukas, Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.
- 3
[IH&MMSEC'21]
AR: 40.3% (128/318)

On the Robustness of Backdoor-based Watermarking in Deep Neural Networks
Masoumeh Shafieinejad, **Nils Lukas**, Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.
- 2
[ACSAC'20]
AR: 20.9% (104/497)

Practical Over-Threshold Multi-Party Private Set Intersection
Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, **Nils Lukas**, John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020.

¹ **Differentially Private Two-Party Set Operations**
[EuroS&P'20] Bailey Kacsmar, Basit Khurram, **Nils Lukas**, Alexander Norton,
AR: 20.9% (39/187) Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri,
Simon Oya, Florian Kerschbaum. IEEE European Symposium on
Security and Privacy (EuroS&P), 2020.

PostDocs, Students

Postdoctoral Researchers

- [Samuele Poppi](#) since 2025
- [Kshitij Mishra](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025
- [Dushyant Chauhan](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025
- [Yuhan Liu](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025

Research Engineers

- [Sarim Hashmi](#) since 2025

PhD Students

- [Fabian Zhang](#) (ML) since 2025
- [Tair Djanibekov](#) (NLP) since 2025
- [Rushil Thareja](#) (NLP) (*secondary: [Praneeth Vepakomma](#)*) since 2024
- [Toluwani Aremu](#) (ML) since 2023

MSc Students

- [Juan Nicolas Sepulveda Arias](#) (CV) since 2025
- [Mohamed Mahmoud Mohamed Hendy](#) (ML) since 2025
- [Majid Mohamed Mohamed Sulaiman Ibrahim](#) (ML) since 2025
- [Prince Jha](#) (ML) (*primary: [Salem Lahlou](#)*) since 2025
- [Muhammad Mohideen](#) (CV) (*secondary: [Samuel Horvath](#)*) since 2025

Alumni

- [Tameem Bakr](#) (ML) MSc · Mar 2026
- [Shaikha Jasem Mohamed Ismail Alhosani](#) (ML) MSc · Mar 2026
- [Maryam Abdulla Rashed Abdulla Alshamsi](#) (ML) MSc · Mar 2026
- [Maryam Alshehyari](#) (ML) MSc · Mar 2026
- [Abdalla Khalid Mohamed Abdalla Alzaabi](#) (ML) MSc · Mar 2026
- [Ali Al-Ali](#) (ML) MSc · Mar 2026

Talks & Keynotes

AI Safety Workshop: Alignment, Oversight and Deception

- [ADIA Lab & UGR Summer School](#), Granada 2026

The Role of Watermarking for ML Security

- [CyberQ](#), UAE Cyber Security Council & TII, Abu Dhabi 2025

Adaptively Robust and Forgery-Resistant Watermarking

- Meta (FAIR), hosted by [Hady Elsahar](#) 2025
- International Symposium on Trustworthy Foundation Models, MBZUAI 2025

Emerging Topics in Machine Learning

- [Build with AI](#), Google Developer Groups, Abu Dhabi 2025

Optimizing Adaptive Attacks against Content Watermarks

- DeepMind, hosted by [David Stutz](#) 2024
- University of California, Berkeley, hosted by [Dawn Song](#) 2024

Analyzing Leakage of Personal Information in Language Models

- Microsoft M365, hosted by [Robert Sim](#) 2024
- Meta, hosted by [Will Bullock](#) 2023
- MongoDB, hosted by [Marilyn George](#) and [Archita Agarwal](#) 2023

How Reliable is Watermarking for Image Generators?

- Google, hosted by [Somesh Jha](#) 2023
- University of California, Berkeley, hosted by [Dawn Song](#) 2023

	Keynotes	
	• Aviation Future Week , hosted by Emirates, Dubai	2024
	• Cyber Energy Leadership Forum , Abu Dhabi	2024
Honors & Awards	Amazon Research Award [100,000 USD]	2025
	First Place at the NeurIPS'24 Watermarking Competition [4,400 USD]	2024
	First Place at DGE Elite Hackathon, GITEX'24 [10,900 USD]	2024
	Top Mathematics Doctoral Thesis , University of Waterloo [1,080 USD]	2024
	Alumni Gold Medal , One PhD Award Yearly, University of Waterloo	2024
	Best Poster Award, Sponsored by David R. Cheriton [220 USD]	2023
	Distinguished Contribution Award, Microsoft MLADS conference	2023
	David R. Cheriton Scholarship, University of Waterloo [14,400 USD]	2022, 2023
	Outstanding Reviewer, ICML'22	2022
	Best Poster Award, Sponsored by Rogers [720 USD]	2019
Workshop Papers	³ [GenAI4Health]	Sanitizing Medical Documents with Differential Privacy using Large Language Models Rushil Thareja, Gautam Gupta, Preslav Nakov, Praneeth Vepakomma, Nils Lukas . 2025.
	² [WMARK'25]	First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.
	¹ [WMARK'25] ★ Oral Presentation	Optimizing Adaptive Attacks against Content Watermarks for Language Models Abdulrahman Diaa, Toluwani Aremu, Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.
Working Papers		Mitigating Watermark Forgery in Generative Models via Randomized Key Selection , Submitted. Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, Nils Lukas . 2025.
		Robust and Calibrated Detection of Authentic Multimedia Content , Submitted. Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, Nils Lukas . 2025.
Journal Publications	² [IEEE S&P'23]	Privacy-Preserving Machine Learning [Cryptography] Florian Kerschbaum, Nils Lukas . IEEE Security & Privacy, Volume 21, Issue 6, 2023.
	¹ [AIP'18]	SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization Pascal Richter, Gregor Heimig, Nils Lukas , Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.
Service	Track Chair	
	• ACM Conference on Computer and Communications Security (CCS), Security and Privacy of Machine Learning (co-chair)	2027
	Area Chair	
	• AAAI Conference on Artificial Intelligence (AAAI), Senior Program Committee	2027
	• International Conference on Learning Representations (ICLR)	2026, 2027
	• Neural Information Processing Systems (NeurIPS)	2026
	• International Conference on Machine Learning (ICML)	2026

Program Committee

- ACM Conference on Computer and Communications Security (CCS) 2025
- IEEE Symposium on Security and Privacy (IEEE S&P) 2025–2027
- Recent Advances in Intrusion Detection (RAID) 2024

Artifact Evaluation Committee

- The ACM Conference on Computer and Communications Security (CCS) 2023, 2024

Reviewer

- European Conference on Computer Vision (ECCV) 2026
- Conference on Language Modeling (COLM) 2026
- British Machine Vision Conference (BMVC) 2026
- FAGEN Workshop @ ICML 2026
- Workshop on GenAI Watermarking (WMark @ ICLR) 2025
- NETYS 2025
- ACM TheWebConf (WWW) 2025
- International Conference on Learning Representations (ICLR) 2024, 2025
- International World Wide Web Conference (TheWebConf) 2024
- Recent Advances in Intrusion Detection (RAID) 2023
- Neural Information Processing Systems (NeurIPS) 2022, 2023
- International Conference on Machine Learning (ICML) 2022, 2025
- The Conference on Information and Knowledge Management (CIKM) 2020

Journal Reviewer

- Communications of the ACM (CACM) 2025

Other

- External Examiner, PhD Dissertation, CISPA Helmholtz Center for Information Security 2026
- Faculty Search Committee, Machine Learning Department at MBZUAI 2025–2026
- Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI 2025
- Admission's Committee, MBZUAI Machine Learning Department 2025
- Student Board Member, Cybersecurity and Privacy Institute 2022–2024
- Session Chair, IEEE Symposium on Security and Privacy (S&P) 2023
- Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS) 2021–2023
- School Advisory Committee on Appointments Liaison, CrySP Lab 2022
- Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA) 2018

Teaching

Instructor, MBZUAI, UAE

- ML8502: Machine Learning Security (14 weeks) 2025, 2026
- ML8509: Collaborative Machine Learning (w/ S. Horvath) 2026
- ML807: Federated Learning (7 weeks) 2025
- ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks) 2024

Teaching Assistant, University of Waterloo, Canada

- CS458/658: Computer Security and Privacy 2020, 2021
- CS246 – Object Oriented Programming 2021

Co-Instructor, RWTH-Aachen, Germany

- Course: Data-driven Medicine 2018

Education

University of Waterloo, Canada 2019 – 02/2024

Ph.D. in Computer Science

- Advisor: [Florian Kerschbaum](#)
- Thesis: Analyzing Threats of Large-Scale Machine Learning Systems

- Thesis Awards: [Top Mathematics Doctoral Prize](#) & [Alumni Gold Medal](#)

RWTH-Aachen, Germany

M.Sc. in Computer Science (*w/Distinction*)

2016 – 2018

B.Sc. in Computer Science

10/2012 – 2016

Work Experience

Assistant Professor, MBZUAI, Abu Dhabi, UAE

since 08/2024

Research Intern, Royal Bank of Canada, Borealis AI, Toronto

2024

- Vertical Federated Learning, hosted by [Kevin Wilson](#)

Research Intern, Microsoft Research, Cambridge, UK

2022

- Privacy for Language Models, hosted by [Shruti Tople](#) & [Lukas Wutschitz](#)

Research Assistant, RWTH-Aachen, Aachen

2014 – 2018

Student Researcher, DSA Daten- und Systemtechnik GmbH, Aachen

2016

Software Engineer Intern, A.R. Bayer DSP Systeme GmbH, Düsseldorf

2012